INCH-POUND

MIL-I-49453(CR)

6 November 1989

MILITARY SPECIFICATION

IMAGE INTENSIFIER ASSEMBLY, 18 MM MICROCHANNEL WAFER MX-10130()/UV

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Image Intensifier Assembly, 18mm Microchannel Wafer, MX-10130A/UV, and the MX-10130B/UV (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards. The following specifications, standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.f).

SPECIFICATIONS

MILITARY

MIL-P-11268 - Parts, Materials, and Processes Used in Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-M-13231	-	Marking of Electronic Items
MIL-B-49030/6	-	Battery, Dry, Type BA-3058/U
MIL-B-49430/4	-	Battery, Primary Lithium, Sulfur Dioxide BA-5567/U
MIL-P-49452	-	Power Supply, AN/PVS-7 Night Vision Goggle, 18mm Microchannel Wafer, Third Generation

STANDARDS

MILITARY

MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-781	-	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

A3140850	-	Lens Assembly Objective, Night Vision Goggle, AN/PVS-7A
A3144305	-	Lens Assembly, F - 1.17 AN/PVS-7B
A3144308	-	Image Intensifier Assembly, 18mm MX-10130B/UV
A3140861	-	Image Intensifier Assembly, 18mm MX-10130A/UV

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Image Intensifier Assembly, 18mm Microchannel Wafer, MX-10130 ()/UV, herein referenced to as the assembly, shall have a minimum useful photocathode (see 3.10.1) and phosphor screen diameter of no less than 17.5 millimeters (mm). The assembly shall employ a Group III-V compound, semi-transparent photocathode sealed to a type 7056 glass faceplate forming the input window to the assembly.

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The assembly shall employ a filmed, microchannel, electron multiplier plate (MCP) with proximity focus on the input and output. The image tube shall contain a noninverting fiber optic screen as an integral part of the tube envelope. The power supply shall include a high-voltage multiplier and oscillator and shall be encapsulated within a hard-surface insulating sleeve or boot.

3.2 Construction. The assembly shall be fabricated in accordance with drawings A3140861 (MX-10130A/UV), A3144308 (MX-10130B/UV) and as specified herein.

3.2.1 Weight. The weight of the assembly shall not exceed 98 grams.

3.3 Qualification. Tubes furnished under this specification shall be product which are authorized by the Government as qualified products (see 3.10.19).

3.3.1 Initial production testing. When specified in the contract, the contractor shall furnish IPT assemblies in accordance with 4.4 (see 6.2.b).

3.4 Materials, parts and processes. Materials, parts, and processes shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor, shall be subject to all provisions of this specification, and shall conform to MIL-P-11268.

3.4.1 Phosphor screen. The phosphor screen shall be type P20 (see 3.10.2). The phosphor material shall be deposited on a suitably metallized fiber optic.

3.5 Components. The following requirements must be met on each Image tube delivered.

3.5.1 Power supply assembly. The power supply assembly (Figure 1) shall meet all requirements of MIL-P-49452.

3.5.2 Fiber optic.

3.5.2.1 Shear distortion. Shear distortion (see 3.10.3) shall not exceed 61 micrometers.

3.5.2.2 Gross distortion. Gross distortion shall cause no point on the image of a straight line to be displaced by more than 75 micrometers relative to the straight line best fitted to the image.

3.5.2.3 Chicken wire. When the fiber optic is uniformly illuminated and viewed under 10-power magnification perpendicular to the plane surface, the fiber optic shall meet the chicken wire (see 3.10.4) limits specified in Table I. Areas in question shall be inspected in such a manner that light is transmitted through the optic so the areas in questions can be observed and measured against the requirements of Table I.

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TABLE I. Fiber Optic Inspection.

Number of allowable incidences	Minimum length	Maximum length
Chicken wire (Zone 1)		
0	0.090	0.090 or greater
2	0.041	0.089
6	0.020	0.040
Disregard lengths which are not greater than 0.019 inch.		
Chicken wire (Zone 2)		
0	0.250	0.250 or greater
5	0.125	0.249
Disregard lengths which are not greater than 0.124 inch.		

Zone 1 - 0.295 inch diameter circle in center of optic
Zone 2 - 0.295 to -.710 inch annulus concentric with Zone 1

3.5.2.4 Surface finish. The output surface of the fiber optic shall have a 60/40 finish over the full useful area of tube (see 3.10.21).

3.5.3 Input Faceplate. The input faceplate shall be 7056 glass of optical quality Grade A with a 60/40 finish and flat within 10 fringes using the sodium D line.

3.5.4 Power source. The tube assembly shall be operational over the input voltage range of 2.0 to 3.0 volts direct current (vdc), using batteries type BA-3058/U or BA-5567/U in accordance with MIL-B-49030/6 or MIL-B-49430/4.

3.5.5 Ion barrier film quality.

3.5.5.1 Type-B hole defects. There shall be no more than 30 type-B (see 3.10.20) holes over the total useful area of the output screen.

3.6 Operational and environmental.

3.6.1 Photocathode sensitivity. The luminous transmission sensitivity shall be not less than 800 microamperes per lumen with a voltage not greater than 800 Vdc applied to the photocathode. The radiant sensitivity shall be not less than 40 milliamperes/watt at 0.8800, $\pm$ 0.0010 micrometer with a voltage not greater than 800 Vdc applied to the photocathode.

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3.6.2 Burn-in/ESS. The assembly shall undergo 50 continuous cycles. Each cycle shall consist of 55 minutes operation and 5 minutes OFF time. Once during each operating cycle the photocathode shall be illuminated with 5×10^{-4} footcandle (fc) for 5 seconds and then with 5 fc for 3 seconds. The remaining portion of the operating cycle shall be without photocathode illumination. The brightness gain setting shall not be changed during or subsequent to completion of burn-in.

3.6.3 Vibration. The assembly with no operating voltage applied shall not be damaged (see 3.10.9) or suffer degradation of performance (see 3.10.23) when subjected to simple harmonic motion parallel to and perpendicular to the optical axis over a frequency range of 5 to 55 Hertz (Hz) with an amplitude of not less than 0.10 inch total excursion for a period of 10 minutes in each plane.

3.6.4 Shock. The operating assembly with no radiation incident on the photocathode shall not be damaged (see 3.10.9) during more than 2 shocks in each direction when subjected to 6 shock impacts parallel to the optical axis and 6 shock impacts perpendicular to the optical axis. Impacts shall be half-sinewave with a minimum peak amplitude of 75 g's (see 3.10.5) and a duration of 6, $\pm$ 2 milliseconds measured at the 10 percent amplitude points.

3.6.5 Environmental temperature. The assembly shall not be damaged by storage, operation or thermal shock when subjected to the environmental temperature profile specified in Figure 2 and shall meet the following requirements specified.

3.6.5.1 Temperature +49°C (+120°F).

a. Input current - Not greater than 35 milliamperes.

b. Gain at 2×10^{-6} footcandle - Not less than 10,000 and not greater than 50,000.

c. Gain at 2×10^{-4} footcandle - Not less than 3,500 and not greater than 10,500.

d. Operational stability:

(1) The output brightness shall not fluctuate more than $\pm$ 10 percent from a steady-state condition and shall not drift more than $\pm$ 15 percent from a steady-state condition for a period of 2 minutes after an initial 2 seconds of operating time. All fluctuations shall be random after the initial 2 seconds.

(2) If random fluctuations greater than $\pm$ 10 percent but less than $\pm$ 15 percent occur in the original 2-minute time period, the test shall be continued an additional 2 minutes, during which the requirements of (1) shall be met.

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(3) If the output brightness drift is greater than ± 15 percent during the original 2-minute time period, the test shall be continued an additional 2 minutes during which the requirements of (1) shall be met.

e. Rise time - shall be not greater than 7 seconds. Overshoot shall be not greater than 40 percent of steady-state output brightness (See 3.10.18).

3.6.5.2 Temperature -51°C (-60°F).

a. Input current - Not greater than 25 milliamperes.

b. Gain at 2 x 10^{-6} footcandle - Not less than 10,000 and greater than 50,000.

c. Gain at 2 x 10^{-4} footcandle - Not less than 3,500 and greater than 10,500.

d. Operational stability:

(1) The output brightness shall not fluctuate more than ± 10 percent from a steady-state condition and shall not drift more than ± 15 percent from a steady-state condition for a period of 2 minutes after an initial 2 seconds of operating time. All fluctuations shall be random, after the initial 2 seconds.

(2) If random fluctuations greater than ± 10 percent but less than ± 15 percent during the original 2 minute time period, the test shall be continued an additional 2 minutes during which the requirements of (1) shall be met.

(3) If the output brightness drift is greater than ± 15 percent during the original 2-minute time period, the test shall be continued an additional 2 minutes during which the requirements of (1) shall be met.

e. Rise time - shall be not greater than 7 seconds. Overshoot shall be less than 40 percent of steady-state output brightness.

f. After exposure of the assembly to the profile of Figure 2, the room temperature (see 3.10.7) performance shall be as follows:

(1) Input current - Not greater than 30 milliamperes.

(2) Gain at 2 x 10^{-6} footcandle - Not less than 20,000 and not greater than 35,000.

(3) Gain at 2 x 10^{-4} footcandle - Not less than 3,500 and not greater than 10,500.

(4) Operational stability:

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(a) The output brightness shall not fluctuate more than ± 10 percent from a steady-state condition and shall not drift more than ± 15 percent from a steady-state condition for a period of 2 minutes after an initial 2 seconds of operating time. All fluctuations shall be random after the initial 2 seconds.

(b) If random fluctuations greater than ± 10 percent but less than ± 15 percent occur in the original 2-minute time period, the test shall be continued an additional 2 minutes during which the requirements of (a) shall be met.

(c) If the output brightness drift is greater than ± 15 percent during the original 2-minute period, the test shall be continued an additional 2 minutes during which the requirements of (a) shall be met.

(d) The variation between the room temperature gain tested prior to high temperature and after low temperature (post test) shall not be greater than ± 15 percent. Room temperature EBI after high and low temperature tests shall be as specified in 3.6.6.

3.6.5.3 Humidity test. The unprotected tube assembly shall not be damaged (see 3.10.9) when subjected to the humidity requirement specified in MIL-STD-810, Method 507.1, Procedure II. Gain may vary up to ± 20% and EBI is allowed to vary up to 3X the value during humidity testing. All parameters must remain in specification limits.

3.6.6 Equivalent background input (EBI). The equivalent background input at room temperature shall not exceed 2.5×10^{-11} phot (lumens/square centimeter) when tested at 23°C ± 2°C.

3.6.7 Luminance gain. The assembly shall have a room temperature luminance gain and high light level saturation characteristics as specified in Table II and Figure 3.

TABLE II. Saturation requirements for Luminance gain.

Nominal light input (fc)	Minimum allowable gain	Minimum allowable output (fl)	Maximum allowable gain	Maximum allowable Output (fl)	Input current (mA) max
2×10^{-6}	20,000	NA	35,000	NA	30
2×10^{-4}	3,500	NA	10,500	NA	30
1.0	NA	0.7	NA	2.1	NA
20.0	NA	0.7	NA	2.1	NA

3.6.8 Halo. The halo (see 3.10.8) produced by projecting a spot of light onto the input of the assembly shall be no greater than 1.47 mm in diameter.

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3.6.9 Bright source protection. The assembly shall not be damaged (see 3.10.9) when subjected to an input illumination of not less than 50 millilumens concentrated on the photocathode within an area no greater than 1 square millimeter for a time interval not less than 1 minute. There shall no discernible damage after a non-operating period of not more than 24 hours, under the test conditions of 4.6.18.

3.6.10 Signal-to-noise ratio. The signal-to-noise ratio shall be not less than 14.5.

3.6.11 Fixed pattern noise.

3.6.11.1 Multi-to-multi pattern variation (see 3.10.10). Multi-to-multi brightness deviations from the mean value shall not exceed ± 10 percent.

3.6.11.2 Multi-boundary pattern noise (see 3.10.11). The average value of the brightness deviations of the multi-boundary intensities shall not deviate from the mean value of the adjacent multi intensities by more than ± 10 percent. The mean value shall be established from three adjacent multies containing the above multi-boundaries.

3.6.12 Output brightness uniformity. When the photocathode is uniformly illuminated with light at a color temperature of 2856, ± 50K, the output brightness uniformity shall be such that the ratio of the maximum to minimum brightness variation over the useful screen area shall not exceed 3:1. For input illumination of wavelength 0.880 ± 0.0010 micrometer, the ratio shall not exceed 4:1. Under the same conditions, when the screen is viewed with a 10-power magnifier, the background shading shall be uniformly graded with no distinct lines of demarcation between the light and dark areas.

3.6.13 Veiling glare. The evidence of chrome visible between a cathode and the black glass of the input faceplate shall not exceed a veiling glare standard by greater than 10% (percent).

3.6.14 Resolution.

3.6.14.1 Center resolution. The center resolution, referenced to the photocathode, shall be not less than 36 line pairs per millimeter (lp/mm).

3.6.14.2 Peripheral resolution. The peripheral resolution, referenced to the photocathode, shall be not less than 36 lp/mm. This requirement shall be met at two points separated by 90 degrees spaced on a 14 millimeter diameter circle concentric with the optical axis (see 3.10.15).

3.6.14.3 High-light-level center resolution. With the entire photocathode of the assembly illuminated with not less than 20 footcandles, the minimum center resolution shall be 5 lp/mm. This requirement shall be performed during Group A and Group D testing only.

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3.6.15 Modulation transfer function (see 3.10.12). The minimum assembly modulation transfer function (MTF) shall be as follows:

a. 83 percent MTF at 2.5 lp/mm
b. 58 percent MTF at 7.5 lp/mm
c. 20 percent MTF at 15.0 lp/mm

3.6.16 Useful cathode diameter. The useful cathode diameter shall be not less than 17.5 millimeters.

3.6.17 Reversed polarity. The assembly shall not be damaged when subjected to a reversed polarity of 3.0 Vdc input voltage for a period of not less than 60 seconds.

3.6.18 Photocathode, microchannel plate, and screen quality. When the screen is viewed with a 10 power magnifier and with 2 x 10^{-6} footcandle incident on the photocathode, there shall be no discernible field emission brighter than the average background scintillation (see 3.10.13). View the screen with a 10-power magnifier and adjust the radiation level on the photocathode to obtain the best spot contrast. The combined number of dark spots which exceed a contrast (see 3.10.24) of 30 percent of their surrounding areas shall not exceed the sizes and quantities specified in Table III. The areas of noncircular spots shall be determined and the diameters of equivalent circular spots computed. These diameters shall be used to determine the size of noncircular spots when using Table III. When the distance between one or more spots and a given spot is less than the maximum dimension of the given spot, all the above spots shall be considered as one circular spot whose area is equal to that of the smallest rectangle or circle (whichever has least area) which encompasses the above spots and the separation between them. Graininess caused by grainy or "peppery" phosphor screen, channel-to-channel gain variations, or photocathode irregularities (see 3.10.22) resulting in non-uniformities, a dirty window effect (see 3.10.26), shall not be discernible over the useful diameter to the degree that it detracts from normal operation, when viewed with a 10-power magnifier and with the photocathode uniformly illuminated. No bright spots are allowed except those identified as type B holes per paragraph 3.5.5.1.

TABLE III. Tube Assembly dark spots.

Size of spots (inches) From - To	Number of spots within 0.22 inch diameter	Number of spots within annulus bounded by 2 circles 0.22 and 0.58 inch diameter	Number of spots within annulus bounded by 2 circles 0.58 inch diameter and total screen diameter
0.015 -	0	0	0
0.012 - 0.015	0	1	2
0.009 - 0.012	0	5	8
0.006 - 0.009	1	9	23
0.003 - 0.006	3	35	35

NOTE: The 0.22- and 0.58-inch circles shall be concentric with the optical axis of the assembly.

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3.6.19 Luminance persistence. When the light input is gated off, the output brightness of the assembly shall fall to not greater than 0.15 percent within 300 milliseconds after illumination at a level of 4×10^{-3} footcandle.

3.6.20 Image alignment. A test reticle projected on the photocathode of the assembly concentric with the optical axis shall produce an image on the screen of the assembly such that the center of the reticle's image shall fall within a circle 0.006 inch in diameter concentric with the optical axis of the assembly.

3.6.21 EMI. The assembly properly incorporated within the appropriate AN/PVS-7 A or B system shall comply with the EMI requirements of the appropriate system. The requirement tests are RE02 (Radiated Emission) RS02 (Susceptibility Magnetic Induction Field) and RS03 (Susceptibility Electric Field).

3.7 Reliability. The assembly shall have a minimum mean time to failure of not less than 7,500 hours when operated under the standard reliability conditions. The assembly shall have a mean time to failure of not less than 1,500 hours when operated under the accelerated reliability conditions.

3.8 Identification and marking. Each assembly shall be identified and serialized in accordance with MIL-M-13231. The markings shall include a coded acceptance date. The first two numbers shall be the last two digits of the year. The last two numbers of the code shall be two digits indicating the calendar week of the year (01 through 52). Reading from left to right or top to bottom, the code number shall indicate the year and week of acceptance in that order.

3.9 Workmanship. The assembly shall conform to MIL-STD-454, Requirement 9.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.10.1 Photocathode. The glass input window with the photocathode bonded to it is referred to as the photocathode.

3.10.2 Phosphor. The phosphor shall be type P20 as defined commercially.

3.10.3 Shear distortion. Defined as the displacement or rotation of a small line segment with respect to a perfect straight line image.

3.10.4 Chicken wire. Defined as a predominant pattern of dead fibers which has a diameter not greater than 0.0009 inch (2 single fibers) whose light transmission is so degraded that with light projected through the optic, single fibers in the area in question cannot be distinguished or identified as single fibers with the use of 50-power magnification.

3.10.5 "g". g is defined as an acceleration or deceleration of 32.17 ft/s^2.

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3.10.6 Rise time and decay time. The rise time required for the assembly to achieve 50 percent if its steady-state performance after the voltage source is connected to the assembly. The decay time is the time required for the assembly to achieve a screen brightness not greater than 1 x 10^{-3} footlambert after the voltage source is disconnected.

3.10.7 Room temperature. Room temperature is defined as +23°C, +10°C, -2°C for the purpose of this specification.

3.10.8 Halo. Halo is defined as a circular area of brightness evidenced on the assembly output imaging screen occurring as a result of a small bright source input and concentric with the input.

3.10.9 Damage. Damage is defined as:

a. Electrical failure or malfunctioning intermittent or continuous including arcing, corona, flashing, bright spots, flickering, blinking, or change in input current shall not vary more than 1.0mA and shall not vary more than ± 20 percent during or subsequent to completion of burn-in test and Humidity test.

b. Cracks, breakage, deformation, corrosion, or deterioration of any part or finish, and missing or loose components.

c. Deterioration of image quality including fixed pattern noise, ion noise, spots, or shading which exceeds that allowed by 3.6.11, 3.6.12, or 3.6.18.

3.10.10 Multi-to-multi pattern variation. Discernible spatial gain variation between individual multi-patterns or groups of multi-patterns.

3.10.11 Multi-boundary pattern noise. Discernible spatial gain variation between peripheral and interior channels or groups of channels in a multi-pattern.

3.10.12 Modulation transfer function (MTF). Modulation transfer function is a measure of the degradation of a image as it appears at the output screen of the assembly as correlated to the input pattern which is normalized to 100 percent contrast at a spatial frequency equal to or less than 0.2 line pairs per millimeter.

3.10.13 Field emission. Discernible field emission is voltage-dependent extraneous emission which appears as bright spots or a pattern that may flicker or appear intermittently on the image screen in one general area. Field emission is voltage-dependent and is best observed with a low level of radiation incident on the photocathode.

3.10.14 Scintillations. Bright spots which occur on the image screen randomly in space and time.

3.10.15 Optical axis. The optical axis of the assembly is defined as the centerline coaxial with the front surface of the assembly housing aperture.

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3.10.16 Limiting resolution. Limiting resolution is defined as the smallest resolution pattern for which the observer can see and resolve the number of lines in both the vertical and horizontal test patterns.

3.10.17 Environmental gain computations.

a. G_{htjf} High temperature gain

b. G_{ltji} Low temperature gain

c. B_{hji}- High temperature brightness output

d. B_{hoi}- High temperature background brightness output

e. B_{ljf} Low temperature brightness output

f. B_{loi}- Low temperature background brightness output

g. B_{rjf} Room temperature (chamber) brightness output

h. B_{roi}- Room temperature (chamber) background brightness output

3.10.18 Overshoot. The amplitude of output brightness above a steady-state condition at turn-on.

3.10.19 Qualified product. A product which has successfully met all requirements of an IPT or first article witnessed by the procuring activity.

3.10.20 Type B hole defect. A hole in the ion barrier which exists over a string of channels having an area equal to a circle with diameter not greater than 125 micrometers.

3.10.21 Useful area. The useful photocathode and phosphor screen area shall comprise a circle of diameter not less than 17.5 millimeters centered on the assembly optical axis.

3.10.22 Photocathode irregularities. Imperfections in the photocathode which lead to significantly degraded image quality: A photocathode grown on a host crystal cut more than 0.5 degree off the proper crystallographic axis, which leads to a terraced crystal surface; and hillocks or craters due to foreign inclusions in the crystal growth.

3.10.23 Degradation. Degradation is defined as a significant change in measurable characteristics which results in failure of the assembly to meet specified requirements or indicated that there is an inherent defect in the operating characteristics of the units.

3.10.24 Contrast. Contrast is defined as $\frac{B_{max} - B_{min}}{Bmax + Bmin}$, where B_{max} and Bmin represent the greater and the less respectively of two brightness values.

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3.10.25 Nominal voltage. Unless otherwise specified, voltages applied to bare tubes shall be in accordance with the following nominal voltage settings:

(1) Photocathode (V_1) - Not greater than 900 Vdc below microchannel plate input.
(2) Microchannel plate input (V_2) - 800 ± 200 Vdc below microchannel plate output.
(3) Phosphor screen (V_3) - 5700 ± 300 Vdc above microchannel plate output.

3.10.26 Dirty window. A cosmetic defect in the gallium arsenide photocathode that reduces or impairs the image definition of the tube assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Initial production testing (4.4)
b. Quality conformance inspection (4.5)

4.3 Inspection methods. Unless otherwise specified, all inspections shall be performed in accordance with the test methods specified in 4.6. All test methods wherein latitude is desired in test details due to variability of approach and test facilities require Government approval prior to initiation of any testing. The Government reserves the right to request clarification, explanation, or justification of any test method used. Further, the Government reserves the right to deem the justification inadequate and to require reassessment of any and all aspects of any test procedure. Unless otherwise specified, all tests shall be made at room temperature (see 3.10.7).

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4.4 Initial production testing. Unless otherwise specified in the contract or purchase order, the IPT shall be performed by the contractor. Disposition of IPT samples shall be as specified in the contract or purchase order (see 6.2.b).

4.4.1 Initial production testing sample. Unless otherwise specified, the contractor shall furnish 10 sample assemblies as a IPT sample.

4.4.2 Initial production testing. IPT shall consist of Group A, B, C, and D tests. The presence of one or more defects at A, B, and C tests shall be cause for rejection of that assembly and may be cause for rejection of the IPT assemblies. Failure of 4 or more assemblies in Group D test shall be cause for rejection of all IPT assemblies.

4.4.2.1 Group A inspection. Each of the 10 IPT sample assemblies shall be subjected to the tests specified in Tables IV and V. The first three tests in Table V shall be performed in the order listed.

TABLE IV. Inspection.

Defects	Requirement paragraph
Components and materials missing or not as specified	3.4, 3.5
Weight not as specified	3.2.1
Design not as specified	3.2
Chemical and physical properties not as specified	3.2
Dimensions not as specified	3.2
Color not as specified	3.2
Finish not as specified	3.2, 3.5.2.4, 3.5.3
Identification or special marking missing or illegible	3.8
Workmanship not as specified	3.9

TABLE V. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Photocathode sensitivity	3.6.1	4.6.1
Ion barrier film quality	3.5.5	4.6.20
Burn-in/ESS	3.6.2	4.6.2
EBI	3.6.6	4.6.6
Luminance gain	3.6.7	4.6.7
Photocathode, MCP, and screen quality	3.6.18	4.6.18
Resolution	3.6.14	4.6.14
Signal-to-noise ratio	3.6.10	4.6.10
Chrome inspection	3.6.13	4.6.13.1 4.6.13.2

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4.4.2.2 Group B inspection. Ten sample assemblies which have passed all the tests specified in 4.4.2.1 shall be subjected to the tests specified in Table VI.

TABLE VI. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Fixed pattern noise	3.6.11	4.6.11
Output brightness uniformity	3.6.12	4.6.12
Veiling glare	3.6.13	4.6.13

4.4.2.3 Group C inspection. Ten sample assemblies which have passed all the tests specified in Tables IV, V, and VI shall be subjected to the tests specified in Table VII. Vibration and shock tests may be conducted in any order. Failure of either of these two tests shall be considered a failure of both attributes.

TABLE VII. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Modulation transfer function	3.6.15	4.6.15
Vibration	3.6.3	4.6.3
Shock	3.6.4	4.6.4
Environmental/temperature tests	3.6.5	4.6.5
Halo	3.6.8	4.6.8
Bright source protection	3.6.9	4.6.9
Useful cathode diameter	3.6.16	4.6.16
Reversed polarity	3.6.17	4.6.17
Luminance persistence	3.6.19	4.6.19
Image alignment	3.6.20	4.6.20
EMI	3.6.21	4.6.21
Subgroup 2		
Humidity	3.6.5.3	4.6.5.1

4.4.2.4 Reliability Group D inspection. Ten sample assemblies which have passed all the tests in Tables IV, V, VI, and VII shall be subjected to the tests in Table VIII subgroup 1. Upon completion of the Group D reliability requirement, the samples shall remain on test (for information only) until end of life or 4,000 (accelerated) hours. Samples and data shall then be shipped to project manager for Night Vision and Electro-Optics.

4.4.2.5 EMI Group D inspection. Two samples which have successfully passed on the requirements of Tables IV, V, VI and VII shall be incorporated in a system and tested per Table VIII, subgroup 2.

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TABLE VIII. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1 Reliability	3.7	4.7
Subgroup 2 EMI	3.6.21	4.6.21

4.5 Quality conformance inspection.

4.5.1 Group A inspection. Each unit supplied under contract shall be inspected for conformance to the inspections specified in Tables IV and V. The first three inspections listed in Table V shall be performed in the order listed before any other Table V tests. Discrete lots shall be formed from units that pass this inspection.

4.5.2 Group B inspection. Samples shall be selected from lots which have passed the tests specified in 4.5.1 in accordance with the contract or purchase order and shall be subjected to the tests in Table VI.

4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table VII and shall be performed on lots that have passed Group B inspection. Sample units shall be selected in accordance with the contract or purchase order.

4.5.3.1 Sampling for Group C inspection. Samples shall be selected at random for inspections in Table VII, subgroup I from each lot. The first samples selected shall be from the first quality conformance inspection lot at the start of the contract. For subgroup II additional samples may be selected from the lot for humidity testing which can be conducted in parallel to other Group C testing. For luminance persistence tests, each phosphor lot must be certified by conducting tests on a minimum of four representative tubes. Sampling plans shall be as specified in the contract or purchase order (see 6.2.e).

4.5.3.2 Vibration and shock tests. These vibration and shock tests shall be conducted in succession, in any order, under the Group C inspection. Failure of the vibration tests conducted or failure of the shock test conducted shall be considered a failure of both attributes.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract (see 6.2.d).

4.5.4 Group D inspection. This inspection shall consist of the tests specified in Table VIII and shall be performed on units that have passed Group A inspection. The contract shall specify the reliability test (4.7.1 or 4.7.2) to be run (see 6.2.g). EMI shall be conducted during IFT only unless a relative change in design is made.

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4.5.4.1 Sampling for Group D inspection. For the quality conformance reliability Group D inspections, 2 assemblies shall be selected at random from each months' production from assemblies which have passed Group A inspection. If the production quantity exceeds 200 assemblies in a given month, then an additional sample of two will be taken. Samples will be formed into groups of 4 for Group D testing.

4.5.4.2 Group D failure. Actions required relative to reliability Group D failures shall be as specified in the contract (see 6.2.d).

4.5.4.3 Disposition of Group D inspection assemblies. Upon completion of the first, third and fifth Group D reliability inspections in accordance with 4.5.4.1, samples shall be subjected to additional reliability testing (for information only) until samples reach end of life specification (inability to meet the requirements specified in the reliability failure definition or 4000 (accelerated) hours, whichever comes first.) Samples and data shall then be shipped to Project Manager, Night Vision and Electro-Optics, ATTN: AMCPEO-IEW-NVEO, Fort Belvoir, VA 22060-5677. The balance of the Group D reliability samples shall be refurbished, pass Group A inspection and shipped as normal production.

4.6 Operational and environmental tests. Tests shall be conducted in accordance with a contractor-prepared, Government-approved test plan and as specified herein. Unless otherwise specified, the following conditions shall apply:

a. The color temperature of the radiation source shall fall within the following limits for the indicated tests:

Tolerance: 2856 ± 50 K

Test Paragraphs:

4.6.1 Photocathode sensitivity
4.6.6 Equivalent background input (EBI)
4.6.7 Luminance gain
4.6.10 Signal-to-noise
4.6.12 Output brightness uniformity
4.6.13 Veiling glare
4.7 Reliability

Tolerance: 2700K to 2900K

Test Paragraphs:

4.6.2 Burn-in/ESS
4.6.5 Environmental/temperature tests
4.6.8 Halo
4.6.9 Bright source protection
4.6.11 Fixed pattern noise
4.6.14 Resolution
4.6.15 Modulation transfer function
4.6.16 Useful cathode diameter

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4.6.18 Photocathode, microchannel plate, and screen quality
4.6.19 Luminance persistence
4.6.20 Ion barrier film quality

b. The photometer used for screen brightness measurements shall be a Pritchard Model 1970, Gamma Model 2000 or equal.
c. Output brightness to be any value from 0.1 to 10.0 footlamberts uniformly distributed over an output aperture of not less than 17 millimeters.

(1) Tungsten filament lamp operated in conjunction with an opal glass or integrating sphere such that the color temperature of the radiation emitted from the opal glass or integrating sphere is 2856, ± 50 K.

(2) Opal glass or integrating sphere to produce a uniform, lambertian distribution.

(3) Corning Spectral Filters, Nos. 3-71 and 4-67, or equal at output of opal glass or integrating sphere.

(4) Output brightness to be 0.1 to 10.0 footlamberts uniformly distributed over an output aperture of not less than 17 millimeters.

d. The amount of radiation from the source incident on the photocathode of the assembly shall be the amount specified in the test. The 4.6.10 (Signal-to-noise) radiation level shall be within ± 10 percent of that specified. All other radiation levels may deviate by as much as ± 25 percent of that specified, provided that the radiation values used in associated computations are accurate to within ± 10 percent.
e. Meters used for monitoring lamp current and voltage shall have an accuracy of ± 0.25 percent.
f. Tests shall be performed at room temperature (see 3.10.7).
g. Neutral density filters used in test equipment shall have transmission characteristics within 10 percent of the nominal filter transmission from 0.35 to 1.0 micrometer.
h. All tests on the assembly shall be performed with the input voltage at 3.0 Vdc unless otherwise specified. The tolerance on applied nominal input voltages shall be ± 0.05 Vdc.
i. Test chambers used for environmental temperature tests shall maintain the temperature within ± 2° (3.6°F) of the specified test temperature.

4.6.1 Photocathode sensitivity. The photocathode sensitivity shall be measured prior to assembly of the tube and the power supply. Corrections for leakage and dark current shall be made. The luminous sensitivity shall be measured over a useful 16 millimeter diameter circle with 0.001 lumen of tungsten lamp radiation and with a photocathode voltage not greater than 800 Vdc. The photocurrent (in microamperes) corrected for leakage and dark current, divided by the actual input (lumens) is the 2856K photocathode sensitivity. With the same dc voltage applied as above, insert a 0.880 micrometer filter between the photocathode and the tungsten source.

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The 16 millimeter diameter area on the photocathode shall be illuminated with the filtered radiation at a level between 1 x 10^{-8} to 5 x 10^{-6} watt. Measure the total tube current and subtract the photocathode leakage and dark current. This photocurrent (in microamperes), divided by the actual input radiation (in watts) is the cathode radiant sensitivity at 0.8800 ± 0.0010 micrometer (in microamperes per watt). The 0.8800 ± 0.0010 micrometer filter shall have the following characteristics:

a. Far infrared blocking out to 4 micrometers and ultraviolet (UV) blocking out to 0.3 micrometer.
b. Peak placement wavelength to 0.8800 micrometer, ± 0.0010 micrometer.
c. Bandwidth at the 10 percent points of 0.0125, ± 0.0015 micrometer.
d. Minimum peak transmission of 50 percent.

Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Burn-in/ESS. Each assembly shall be operated as specified in 3.6.2 for 50 continuous cycles. The brightness gain setting shall not be changed during or subsequent to completion of the burn-in test. Evidence of damage shall constitute failure of this test. Failure to meet requirements of 3.6.2 shall also constitute failure of this test.

4.6.3 Vibration. The operating potential shall not be applied to the assembly during vibration testing. Tolerance on specified frequencies shall be ± 2 Hz and tolerance on total excursion shall be ± 0.005 inch. Mount the assemblies rigidly, singly or in groups. Subject the assembly to simple harmonic motion applied in a plane parallel to the optical axis of the tube assembly at a varying frequency of 5 to 55 Hz. Vary the frequency logarithmically from 5 to 55 Hz to 5 Hz in one minute. The amplitude of vibration shall be 0.05 inch (0.10 inch total excursion) in each plane. Repeat this 5 Hz - 55 Hz - 10Hz frequency sweep 10 times. At the conclusion of the 10 frequency sweeps, apply the simple harmonic motion to the assembly in a plane perpendicular to the optical axis of the assembly and repeat the above frequency sweeps. Failure to meet the requirements of 3.6.3 or 4.6.3.1 shall constitute failure of this test.

4.6.3.1 Post tests for vibration. If vibration is conducted after shock the post tests of 3.6.6, 3.6.7, 3.6.10, 3.6.14.1 and 3.6.18 shall be conducted to assure successful completion of both vibration and shock.

4.6.4 Shock. This test shall be conducted in a darkened environment with no light incident on the photocathode and with the operating potential applied. Apply 6 shock impacts both parallel to and perpendicular to the optical axis in such a way as to generate nominal half sine wave pulses having a minimum peak amplitude of 75 g. The duration of each shock pulse shall be 6, ± 2 milliseconds, measured between the 10 percent values of peak amplitudes. The after-oscillations shall be not greater than 15 percent of the peak amplitudes of the nominal half sine wave pulses. Failure to meet requirements of 3.6.4, or 4.6.4.1 shall constitute failure of this test.

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NOTE: There shall be not more that 2 instances of flashing or flickering during the 6 shock impacts on each axis. The horizontal and vertical plane shocks may be performed in either order.

a. Mount the assembly with the optical axis in a vertical plane and subject it to 6 shock impacts with the direction of the force applied parallel to the optical axis.

b. Observe the image screen with the unaided eye during application of shock impacts for evidence of flashing, flickering, bright spots, or electrical breakdown.

c. Mount the assembly with the optical axis in a horizontal plane and subject it to 6 shock impacts with the direction force applied perpendicular to the optical axis.

d. Observe the image screen with the unaided eye during application of shock impacts for evidence of flashing, flickering, bright spots, or electrical breakdown.

4.6.4.1 Post tests for shock. If shock is conducted after vibration the post tests of 3.6.6, 3.6.7, 3.6.10, 3.6.14.1 and 3.6.18 shall be conducted to assure successful completion of both shock and vibration.

4.6.5 Environmental/temperature tests. Place the assembly in a test chamber at room temperature and perform the room temperature tests specified in Table IX. A minimum 18.2 ± 0.1 mm diameter area of the input window shall be illuminated by the specified input illumination. All gain measurements shall be computed using readings referenced to a room temperature luminance gain baseline measured prior to this test in accordance with 4.6.7. Remove the operating potential. Raise the temperature of the test chamber to +71°C, and hold at this temperature for a minimum of 2 hours. At the end of the 2-hour stabilizing period, lower the test chamber temperature to +49°C and hold at this temperature for a minimum of 6 hours. After 6 hours at +49°C, subject the assembly to the high temperature (+49°) performance tests specified in Table IX. The assembly shall be off not less than 3 minutes prior to performance of the rise time test. The rise time shall be recorded on an X-Y recorder, or equivalent. The operational stability of the assembly shall be verified by viewing the output signal from the photometer with a device that will display the signal for the period specified in 3.6.5.1.d. Adjust the vertical sensitivity to obtain a minimum of 70 percent deflection from a zero reference. Apply the specified operating potential to the assembly, and after not greater than 15 seconds illuminate the photocathode with the specified step pulse of input illumination. The turn-on time of the input light pulse shall be not greater than 700 milliseconds from zero brightness to 100 percent brightness. After the +49°C tests are completed, remove the operating potential. Raise the temperature of the test chamber to +71°C and hold at this temperature for not less than 30 minutes. Thermal shock the assembly from +71°C to room temperature within 3 minutes or less. Lower the temperature of the test chamber to -51°C and hold at this temperature for a minimum of 1 hour. After the 1-hour stabilizing period, subject the assembly to the low temperature (-51°C) performance tests specified in Table IX.

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The test procedures for rise time, gain, and operational stability shall be the same as those used at +49°C. At the conclusion of the -51°C tests, remove the operating potential to the assembly. Lower the temperature of the test chamber to -56.6°C and remain at this temperature for not less than 2 hours. At the end of the stabilizing period, thermal shock the assembly from -56.6°C to room temperature within 3 minutes or less. Hold the assembly at room temperature for a minimum of 4 hours. At the end of the room temperature holding period, subject the assembly to the room temperature tests specified in Table IX. The test procedures for operational stability shall be the same as that used at +49°C, except that the light pulse for illuminating the photocathode shall be applied within 10 seconds after the operating potential has been applied to the assembly. Compute the high temperature luminance gain (G_{htji}):

$$G_{htji} = \frac{B_{hji} - B_{hoi}}{B_{rjf} - B_{roi}} \times \text{room temperature gain}$$

Compute the low temperature luminance gain (G_{ltji}):

$$G_{ltji} = \frac{B_{ljf} - B_{loi}}{B_{rjf} - B_{roi}} \times \text{room temperature gain}$$

Failure to meet requirements of 3.6.5, 3.6.6, 3.6.7, 3.6.10, 3.6.14.1, and 3.6.18 shall constitute failure of this test.

TABLE IX. Environmental/Temperature tests.

Temperature	Input light (fc)	Gain	Input current	Rise time		Operational stability	
Input voltage		3.0 Vdc	3.0 Vdc	2.0 Vdc	3.0 Vdc	2.0 Vdc	3.0Vdc
+23°C	2×10^{-6} 2×10^{-4} 1	*(1) *(1)	*(1)			*(1) *(1)	*(1) *(1)
+49°C	2×10^{-6} 2×10^{-4} 1	* *	* (*)	*	(*)	(*) *	(*) (*)
-51°C	2×10^{-6} 2×10^{-4} 1	*	*	*	(*)	* *	(*) (*)

* Applicable
(*) Applicable to IPT only
(1) Test after completion of environmental temperature cycle

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4.6.5.1 Humidity test. Subject the unprotected assemblies to the temperature/humidity profile of MIL-STD-810, Method 507.1, Procedure II as modified below. Failure to meet the requirements of paragraph 3.6.5.3 and the tests specified shall constitute failure of this test.

Procedure II - Ground and airborne electronic equipment.

Step 1 - Delete
Step 2 - Delete
Step 3 - Delete
Step 4 - Take initial measurements of gain, input current, center resolution and EBI. Gain shall be run at 2×10^{-6} fc.

NOTE: No readjustment shall be permitted throughout the test period and no repairs or replacement of parts shall be permitted.

Step 5 - As - is
Step 6 - As - is except test the U.U.T at the end of the third cycle. Test per Step 4 above. Wiping to remove excess moisture is permitted. Changes in gain and input current up to $\pm$ 20 percent is allowed. EBI may change to a value 3 times the start value. No value for any tests may exceed requirements.
Step 7 - At the end of five continuous 48 hours cycle, cool the assemblies to room temperature and hold at room temperature for a maximum of four hours.
Step 8 - At the end of the room temperature holding period, subject the assemblies to the measurements specified in Step 4 Above.
Step 9 - Delete

4.6.6 Equivalent background input (EBI). With the operating potential applied to the assembly and no radiation incident on the photocathode, hold for not less than a two minute period; then measure the screen brightness (footlamberts) with a photometer after the screen brightness has stabilized and record the photometer reading (R_1). Illuminate the photocathode at a level between 3.5 to 7×10^{-11} phot (lumens per square centimeter) uniformly distributed over the full useful diameter centered on the photocathode faceplate. Record photometer reading (R_2). The angle of incident flux shall not be greater than 2 degrees from perpendicular and the photometer shall be positioned such that the acceptance angle covers the full 17 millimeters. Determine the equivalent background input (EBI) by the following formula:

$$EBI = \frac{R_1}{R_2 - R_1} \times \text{actual input illumination}$$

Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.6.6.1 EBI alternate. An alternate method of measuring EBI which may be used in lieu of that specified in 4.6.6 is as follows: With the operating potential applied to the intensifier and no radiation incident on the photocathode hold for a period of not less than 2 minutes. At the end of this period with no radiation incident on the photocathode measure the screen brightness.

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Divide this reading by the luminance gain and multiply the result by 1.076 x 10^{-3}. Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Luminance gain. Illuminate the photocathode of the operating tube assembly with an input light level, uniformly distributed over the full useful diameter of the photocathode faceplate, in accordance with each of the conditions of 3.6.7 in sequence. Measure the input current and measure the image screen luminance with the photometer for each condition. The acceptance angle of the photometer shall be not greater than 2 degrees. Position the photometer so that the acceptance angle subtends a 17mm diameter area centered on the phosphor screen. Luminance gain is determined by dividing the screen luminance in footlamberts by the actual input illumination in footcandles. Failure to meet requirements of 3.6.7 shall constitute failure of this test.

4.6.7.1 Exposure of the operating assembly to a light level resulting in ABC operation prior to the measurement of low light level gain is necessary to ensure that errors resulting from initial overshoot/undershoot are not incorporated into the the gain measurement.

4.6.8 Halo. Illuminate the photocathode of the assembly through an aperture followed by the appropriate AN/PVS-7 objective lens (A or B) or equivalent. The aperture shall be such as to produce a spot of light 0.350, $\pm$ 0.02 mm in diameter on the photocathode of the assembly. The illumination in that spot shall be not less than 5 x 10^{-4} footcandle. The illumination incident on the photocathode of the assembly in the region outside of the 0.350 millimeter spot shall not exceed 5 x 10^{-7} footcandle. The illumination incident on the photocathode of the assembly in the region outside of the 0.350 millimeter spot shall not exceed 5 x 10^{-7} footcandle. Measure the diameter of the halo (see 3.10.8) formed on the output image screen with no less than a 10-power measuring magnifier. Repeat the measurement 3 times and compute an average diameter. Failure to meet requirements of 3.6.8 shall constitute failure of this test.

4.6.8.1 Halo alternate. Illuminate the photocathode of the assembly through two apertures followed by an objective lens. The apertures shall be such as to produce two spots of light 0.350 $\pm$ 0.02mm in diameter with centers displaced by 1.47mm on the photocathode of the assembly. The illumination of the spots shall be not less than 5 x 10^{-4} footcandle. The illumination incident on the photocathode of the assembly in the region outside the 0.350 millimeter spots shall not exceed 5 x 10^{-7} footcandle. The diameter of the Halos formed on the output image screen shall not touch or overlap when viewed with no less than a 10-power magnifier. Failure to meet the requirements of 3.6.8 shall constitute failure of the test.

4.6.9 Bright source protection. Apply the input potential to the assembly. Illuminate the photocathode with a spot of light having an area not greater than 1.0 square millimeter and having an intensity of at least 50 millilumens. Failure to meet requirements of 3.6.9 shall constitute failure of this test.

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4.6.10 Signal-to-noise. Image a circular spot no larger than 0.200 millimeter in diameter onto the photocathode within a 2.0 millimeter diameter circle centered on the assembly optical axis. Tube dark spots (see 3.6.18) may be avoided for spot placement. Any spot movement outside the 2.0 millimeter circle shall require that two separate locations, which are diametrically opposed relative to the optical axis and which are within a 5.6 millimeter circle centered on the optical axis, be measured and averaged to determine signal-to-noise ratio. The circular spot shall uniformly illuminate the photocathode at a level no greater than 1×10^{-5} footcandle. Focus the signal which is emergent from the assembly onto a pinhole 0.200 millimeter or larger in diameter. Align to obtain a maximum signal through the pinhole. Measure the light passing through the pinhole with a low-dark-current photomultiplier tube, EMR Model 541E, or equivalent. After suitable amplification, pass the signal to a Total Technology type, or equivalent signal-to-noise measurement set and measure the dc content and the rms value of the signal over an electronic bandwidth of 10Hz. The signal-to-noise (S/N) ratio is the ratio of the dc signal to the rms noise:

$$S/N = \frac{S_o - S_{bkd}}{K\,(N_o^2 - N_{bkd}^2)^{1/2}}$$

where:

S = dc signal

N = rms noise

S_o = signal output from S/N test set

S_{bkd} = background signal from S/N test set with light input off

N_o = noise output from S/N test set rms meter

N_{bkd} = background noise from S/N set rms meter with light input off

K = correlation factor to obtain a signal-to-noise ratio over an equivalent bandwidth of 10 Hz independent of assembly frequency response;

K = 1.19 for a P20 phosphor.

Failure to meet requirements of 3.6.10 shall constitute failure of this test.

4.6.11 Fixed pattern noise. With the operating potential applied to the assembly, illuminate the entire photocathode uniformly at a level of 1 to 2×10^{-4} footcandle. Observe the phosphor screen with a 10-power magnifier for multi-to-multi pattern variations and multi-boundary noise (webbing). If multi-to-multi pattern variations or multi-boundary pattern noise are observed, perform the following:

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a. Choose an area of the image screen where the multi-to-multi pattern appears most noticeable. Search this area for the most contrasting adjacent multi-bundles and measure each multi-bundle brightness, using an aperture of effective diameter equal to one third the distance from flat-to-flat for hexagonal multies. Failure to meet requirements of 3.6.11.1 shall constitute failure of this test.
b. Choose an area of the image screen where the multi-boundary pattern noise is most noticeable. Scan 3 multies and the corresponding multi-boundaries in this area of the screen with a photometer aperture of 25 micrometers effective diameter. Failure to meet requirements of 3.6.11.2 shall constitute failure of this test.

4.6.12 Output brightness uniformity.

4.6.12.1 Uniformly illuminate the photocathode of the assembly under test at a level between 5×10^{-6} and 1×10^{-5} footcandle of 2856K light. Observe the phosphor screen with a 10-power magnifier for shading. If shading is observed, perform the follow:

Using a photometer whose effective aperture at the output image plane of the assembly is not greater than 2.5mm, make four scans completely across the assembly useful area. Each scan shall pass over the assembly optical axis and shall be rotated 45 degrees from the two adjacent scans. The axis of the scanning aperture shall be parallel to the assembly optical axis for all points along each scan. The brightness of the output image plane shall be measured to within $\pm$ 10 percent over each scan, and the brightness values shall be recorded in a suitable storage device. There shall be no compensation made for nonuniformity caused by the output fiber optic. The four scans shall then be examined as a group, from which the maximum and minimum values shall be obtained to compute the maximum-to-minimum output brightness uniformity ratio. Failure to meet requirements of 3.6.12 shall constitute failure of this test.

4.6.12.2 Uniformly irradiate the photocathode of the assembly under test with 1 to 3 x 10^{-10} watt of 0.88-micrometer radiation. Observe the phosphor screen with a 10-power magnifier for shading. If shading is observed repeat the scan procedure in 4.6.12.1. Failure to meet requirements of 3.6.12 shall constitute failure of this test.

4.6.13 Veiling glare.

4.6.13.1 Chrome inspection. Using the stainless steel cathode holder, remove the cathode from the Nikon comparator and place on the back light fixture using quadriceps. Using an A.O. microscope at 10 X with an overhead microscope light source, inspect the seal surface bevel and the outer bevel for open or closed chips, fractures, deep gouges on the seal surface. LOOK AT THE PERIMETER OF THE GaAS CATHODE, THERE SHALL BE NO CHROME VISIBLE AS VIEWED THROUGH THE CLEAR GLASS PORTION OF THE FACEPLATE. REJECT UNIT IF CHROME IS VISIBLE. Testing shall be per 4.6.13.3.

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4.6.13.2 Chrome inspection (alternate). Each lot of tube assemblies shall be screened 100% for evidence of chrome which is visible between a cathode and the black glass of the input faceplate. In the event chrome is visible, testing for veiling glare shall be per 4.6.13.3.

4.6.13.3 Veiling glare sampling. Select samples for testing per the contract or purchase order (see 6.2.f) from the assemblies that exhibit maximum exposure of chrome as observed with the unaided eye. Subject these samples to the veiling glare test procedure described in paragraph 4.6.13.6. In the event that the veiling glare of any of the samples exceeds the measured value of the calibration standard by 10% all tubes with chrome exposure shall be screened for acceptance.

4.6.13.4 Veiling glare calibration standards. Veiling glare shall be measured using a calibrated standard. The calibration standard shall consist of the appropriate AN/PVS-7 Objective Lens Assembly (A3140850, A3144305) and an MX-10130()/UV tube assembly which has no chrome visible when viewed through the clear glass portion of the faceplate. The combined veiling glare for standard objective lens/image tube combination shall not exceed 2.30%.

a. The Objective Lens Assembly standard shall meet all the performance requirements including environmental as specified in the specification.
b. The MX-10130()/UV Tube Assembly standard using a saturn faceplate with no chrome showing shall meet the following paragraph requirements:

3.4.1 Phosphor screen
3.6.7 EBI
3.6.8 Luminance gain
3.6.11 Signal-to-noise
3.6.13 Output brightness uniformity
3.6.16 Center resolution
3.6.19 Useful cathode diameter
3.6.22 Luminance persistence (per phosphor lot)

4.6.13.5 Veiling glare identified standards. The Objective Lens Assembly and the Tube Assembly which are used as the Veiling Glare Standard shall each be identified by serial number.

4.6.13.6 Veiling glare test. Use a NiTec veiling glare tester (drawing number SD000646) or equivalent described as follows:

A black-hole target is imaged on the central 14mm of 18mm image tube, which results in a 2mm annulus bright area at the image tube output. The target itself overfills the lens 40 degrees FOV by at least 25 degrees, in order to account for glare resulting from outsied FOV light. The output black-hole image thus contains an illumination level proportional to the system's glare. The central 4mm diameter area of the black hole is measured for the glare illumination level. Next, the black hole covered to present uniform illumination over the full FOV. The 4mm probe than measures the high-light level illumination. Glare is then calculated as follows:

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The background illumination level (Bd) shall be measured (or pulled) with no target illumination. The annular target shall be illuminated such that is brightness (B1) is $1.2 \pm 0.2 \times 10^{-4}$ footlamberts outside the dark central, and detector reading (BL) shall be recorded. The dark central area of the annular target shall then be elimated such the target's brightness is (B2) over the entire FOV and the detector's reading (BH) shall be recorded. The tube assembly veiling glare (VG) shall be calculated as follows:

$$VG = \frac{(BL - Bd)\ B2}{(BH - Bd)\ B1} \times 100\%$$

Failure to meet the requirements of 3.6.13 shall constitute failure of the test.

4.6.14 Resolution.

4.6.14.1 Center resolution. Perform this test using a projection system having an f-number not greater than f/1.4 to project the test pattern. A resolving power target having black bars on a clear background (1951 Air Force Resolving Power Test Target). A 10-power magnifier. The resolving power test target shall be focused on the photocathode such that the center of the target is aligned with the optical axis. The input radiation shall be adjusted for best image contrast. The image of the resolving power test target formed on the screen of the assembly shall be observed for limiting resolution (see 3.10.16). Failure to meet requirements of 3.6.14.1 shall constitute failure of this test.

4.6.14.2 Peripheral resolution. Determine the peripheral resolution with the same or equivalent test equipment to that used in 4.6.14.1. The resolving power target shall be focused on the photocathode such that the group and element representing 36 line pairs per millimeter of the target positioned 7 mm from the center of the assembly housing aperture. Input radiation shall be adjusted for best image contrast. Failure to meet requirements of 3.6.14.2 shall constitute failure of this test.

4.6.14.3 High Light level center resolution. The resolving power test target shall be focused on the photocathode such that the center of the target is aligned with the optical axis. The input radiation shall be adjusted to no less than twenty (20) footcandles in accordance with 4.6. The image of the resolving power target found on the screen of the assembly shall be observed for the limiting resolution (3.10.16). Resolution less than that specified in 3.6.14.3 shall constitute failure of this test.

4.6.15 Modulation transfer function (MTF). Equipment used to measure modulation transfer function must satisfy the following, or equivalent, subject to Government approval.

a. A sine wave analyzer capable of, but not limited to, direct readout for spatial frequencies within one line pair of the following: 2.5 lp/mm, 7.5 lp/mm and 15 lp/mm such that MTF determined at each frequency shall be within $\pm$ 2 percent of the true value.

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b. The analyzing slit must be 10 micrometers or less in width by 1 millimeter or more in length.

c. The limiting aperture in the plane of the test pattern shall be a minimum of 2.5 millimeter in width referred to the phosphor screen of the assembly.

d. The test systems calibration frequency shall be 0.2 lp/mm or less.

e. The test systems MTF shall be greater than 95 percent at 2.5 lp/mm.

f. Input illumination measured at the assembly photocathode shall be not greater than 8×10^{-4} fc.

The zero MTF level shall be determined by blocking the light from the phosphor screen while the system is running at calibration frequency. When unblocked, the calibration reading is normalized to 100 percent MTF. Assembly MTF is determined by dividing MTF of the measuring system with the assembly in place by MTF of the measuring system without the assembly. Failure to meet requirements of 3.6.15 shall constitute failure of this test.

4.6.16 Useful cathode diameter. With the operating potential applied to the assembly, focus the test reticle on the photocathode such that it is centered on the assembly's optical axis. The test reticle shall consist of equally spaced lines from the center to the edge of the reticle in four directions 90° apart. Spacing between a large graduation and a small graduation shall be 0.5 millimeter $\pm$ 0.03 millimeter, and spacing between two large graduations shall be 1.0 millimeter $\pm$ 0.03 millimeter. All lines, letters and numbers shall be high contrast black on a clear glass substrate. Adjust the input radiation level for best image contrast. The output screen shall be viewed with a 10-power magnifier. Useful cathode diameter is determined by the number of millimeter graduations visible on the screen of the assembly and shall be determined in both vertical and horizontal directions. Failure to meet requirements of 3.6.16 in both vertical and horizontal directions shall constitute failure of this test.

4.6.17 Reversed polarity. With no light on the photocathode, the assembly shall be operated with a reversed polarity input voltage for a period of not less than 60 seconds. Failure to meet requirements of 3.6.17 shall constitute failure of this test.

4.6.18 Photocathode, microchannel plate, and screen quality. With the operating potential applied and no radiation incident on the photocathode, observe the image screen with a 10-power magnifier or greater. With an input light level adjusted for best spot contrast, the image screen shall be observed for dark spots that exceed the size or quantity specified in Table III. With an input light level of 2×10^{-6} fc present to a degree that it detracts from normal performance, observe the screen for field emission and graininess. If chicken wire is present to a degree that it distracts from normal performance refer to the requirements of 3.5.2 and Table I. Failure to meet requirements of 3.6.18 and 3.5.2.3 shall constitute failure of this test.

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4.6.19 Luminance persistence. Illuminate the central part of the photocathode with a 1/8 inch spot of light at 4 x 10^{-3} fc, ± 10 percent for 1 minute. Let B_s represent the screen brightness. Gate the light off within 1 milliseconds (ms) and observe the output screen brightness, $B_d(t)$, where t is time after gating the light off. The persistence $(B_d(t)/B_s) \times 100$ percent at 300 ms shall not be greater than the values specified in 3.6.19. Failure to meet requirements of 3.6.19 shall constitute failure of this test.

4.6.20 Ion barrier film quality. The ion barrier film quality shall be tested prior to assembly with the power supply. When the photocathode is uniformly illuminated with not less than 1 x 10^{-1} footcandles, apply potentials V_2 and V_3 (3.10.25). Adjust V_1 (3.10.25) between 1 volt and 8 volts for best contrast while observing the phosphor screen with a 10-power eyepiece or microscope of sufficient power to resolve the holes with a dark-adapted eye. V_2 and V_3 shall be those selected nominal (3.10.25) operating voltages for that particular tube under test to obtain a luminance gain of 20,000 - 35,000 FL/FC for 2 x 10^{-6} footcandles illumination on the photocathode. Photograph the output screen with not less than 1 x 10^{-1} footcandles illumination on the photocathode. This photograph shall be of sufficient quality to be representative of the holes as viewed with a 10 power eyepiece and a dark-adapted eye. Failure to meet the requirements of paragraph 3.5.5.1 shall constitute failure of this test.

4.6.21 Electromagnetic interference. The AN/PVS-7A, B system under test shall consist of a monocular assembly powered by an internal battery. Testing shall be in accordance with paragraph 4.6.5 of the associated system specification.

4.6.22 Image alignment. Perform this test with the same equipment or equivalent, used in 4.6.14. With the operating potential applied, focus a test reticle on the photocathode. The photocathode shall be illuminated to provide a high contrast image of the test reticle. Observe the image of the test reticle formed on the screen of the assembly with a 10-power measuring magnifier containing a circular pattern of 0.006-inch diameter. When the image of the center of the test reticle is focused on the optical axis of the photocathode, the screen image of the reticle center shall be examined to verify that it falls within the circular pattern. The magnifier shall be aligned with the optical axis of the assembly such that the center of the circular pattern falls on the optical axis. Failure to meet requirements of 3.6.14 shall constitute failure of this test.

4.6.22.1 Image alignment-alternate. Project a 0.056 ± 0.001 inch spot onto the photocathode. This spot shall fall within a 0.062 ± 0.001 inch output aperture. Failure of the 0.056 ± 0.001 inch spot to fall within the 0.062 ± 0.001 inch spot shall constitute failure of this test.

4.7 Reliability. The tests shall be in accordance with Table XI and as specified herein. Sufficient instrumentation shall be provided to ensure immediate recognition of a catastrophic failure as well as a change in relative output brightness.

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Assembly parameters; luminance gain; EBI; signal-to-noise; center resolution; useful cathode diameter; photocathode, microchannel plate, and screen quality; and output brightness uniformity shall be tested at room temperature as specified herein not less than once each 200 ± 50 hours of operating time. Reliability test assemblies shall be removed from test conditions for not more than 15 hours at any given time for measurement of assembly parameters. At other times, assemblies shall be cycled continuously on a 24 hour basis. Acceptance levels are as specified in Table X. Failure to meet these criteria constitutes failure of that unit. A failure shall be presumed to have occurred immediately after the last successful measurement or inspection. Failed assemblies which have been removed from test shall not be replaced or returned to test unless for information only. The requirements of MIL-STD-781 shall apply. Failures of the power supply shall be considered relevant. The contractor shall determine which reliability test he will conduct; standard or accelerated.

TABLE X. Reliability test acceptance levels.

Inspection	Acceptance level
Luminance gain (at 2 X 10^{-6} fc)	8,000 to 35,000
EBI	2.5 X 10^{-11} phot, maximum
Signal-to-noise	9.0 minimum
Center resolution	32 lp/mm, minimum
Useful cathode diameter	17.5 mm, minimum
Output brightness uniformity At 2856 ± 50 degrees K At 0.88 um	 3:1 maximum 4:1 maximum
Image quality	As per 3.6.18 and 4.6.18

TABLE XI. Reliability test environmental conditions.

Parameter	Environmental conditions
Temperature	40°C, ± 5°C
Temperature cycling	Not applicable
Vibration	2.2 g, ± 10% peak acceleration at any nonresonant frequency between 20 and 60 Hz, measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each period of equipment ON time when performing standard reliability. The duration of vibration shall be at least 2 minutes during each period of equipment ON time when performing accelerated reliability.

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4.7.1 Standard reliability tests. ON-OFF cycling shall consist of 55 minutes of operation (ON time) followed by 5 minutes OFF. Input voltage to the assembly shall be 2.7 to 3.0 Vdc. The initial brightness gain shall be between 20,000 and 35,000. No gain adjustments shall be allowed during the test or inspections. During operation each assembly shall be illuminated with not less than 1×10^{-5} fc. Once during each 55 minutes of operation the illumination of each assembly shall be increased to 5×10^{-4} fc for 5 seconds, reduced to 1×10^{-5} fc for not less than 5 minutes, and then increased to 5 fc for 3 seconds. See 4.7.3 for failure definition.

4.7.2 Accelerated reliability test. ON-OFF cycling shall consist of 11 minutes of operation (ON time) followed by 1 minutes OFF. Input voltage to the assembly shall be 2.7 to 3.0 Vdc. The initial brightness gain of each assembly shall be between 20,000 and 35,000. No gain adjustments shall be allowed during this test or inspection. During operation each assembly shall be illuminated with not less than 5×10^{-5} fc. Once during each 11 minutes of operation the illumination of each assembly shall be increased to 5×10^{-4} fc for 5 seconds, reduced to 5×10^{-5} fc for not less than 5 minutes and then increased to 5 fc for not less than 3 seconds. See 4.7.3 for failure definition.

4.7.3 Accept/reject criteria.

4.7.3.1 Initial production test reliability inspection. The accept/reject determination shall be made upon either the accumulation of 7500 hours operating time (1500 hours accelerated) on each of the ten assemblies or the occurrence of four assembly failures. Four or more assembly failures shall constitute failure of the equipment to meet the reliability inspection requirement. Three or less assembly failures shall constitute acceptance of the reliability requirement.

4.7.3.2 Quality conformance reliability inspection. The accept/reject determination shall be made upon either the accumulation of 7500 hours operation time (1500 hours accelerated) on each of the four assemblies or the occurrence of two assembly failures. Two or more assembly failures shall constitute failure of the equipment to meet the reliability inspection requirement. One or less assembly failures shall constitute acceptance of the reliability requirement.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The assembly is intended for use in the AN/PVS-7 Night Vision Goggle, an electro-optical viewing device capable of intensifying low light levels such that a visible image is presented for viewing and sighting purposes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Time frame required for submission of IPT number of assemblies required and disposition of IPT samples (see 4.4).
c. Disposition of Group C and D assemblies.
d. Necessary action by contractor in event of Group C or D failures.
e. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling should be conducted per the requirements of MIL-STD-105.
f. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
g. Qualification - If product is not qualified at time of award the contract must require qualification prior to first delivery.
h. MIL-STD-810C should be used for environmental tests in section 4.

6.3 Definitions. See 3.10.

6.4 Subject term keyword listing.
Image Intensifier Assembly
Microchannel Wafer

Custodian:
Army - CR

Preparing Activity:
Army - CR

Project 5855-A318

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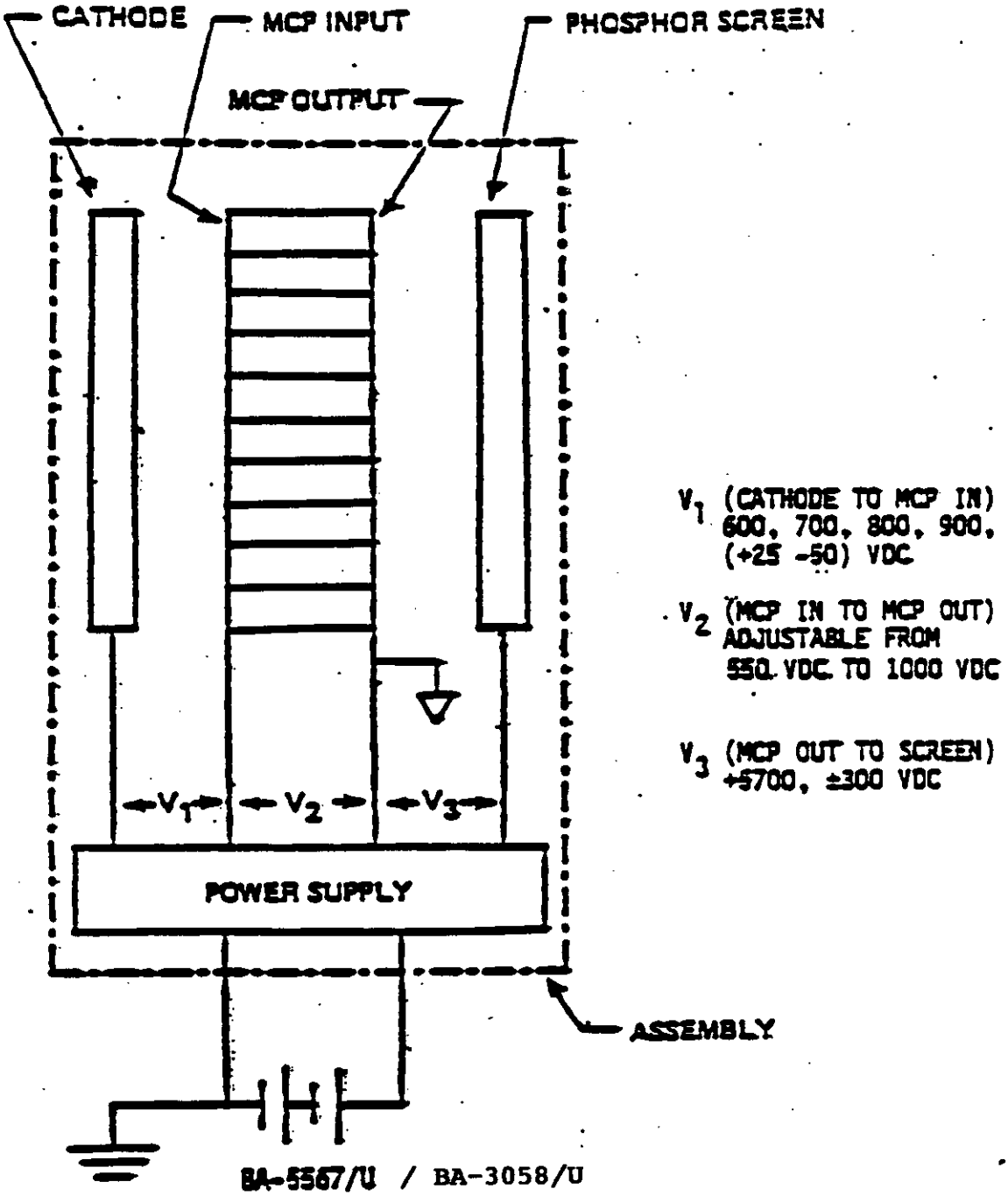


FIGURE 1. Power supply/image tube interface.

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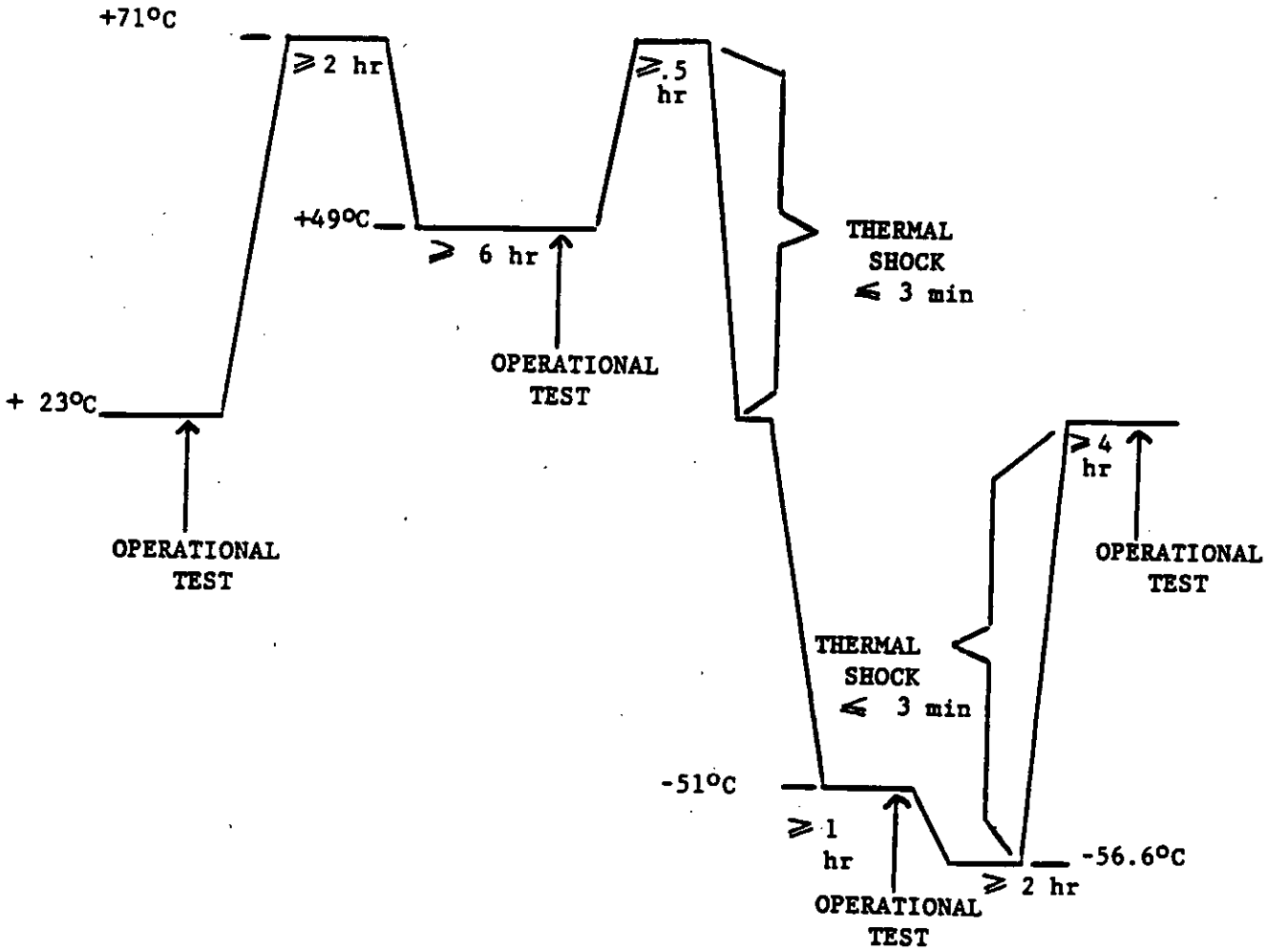


FIGURE 2. Environmental Temperature Profile

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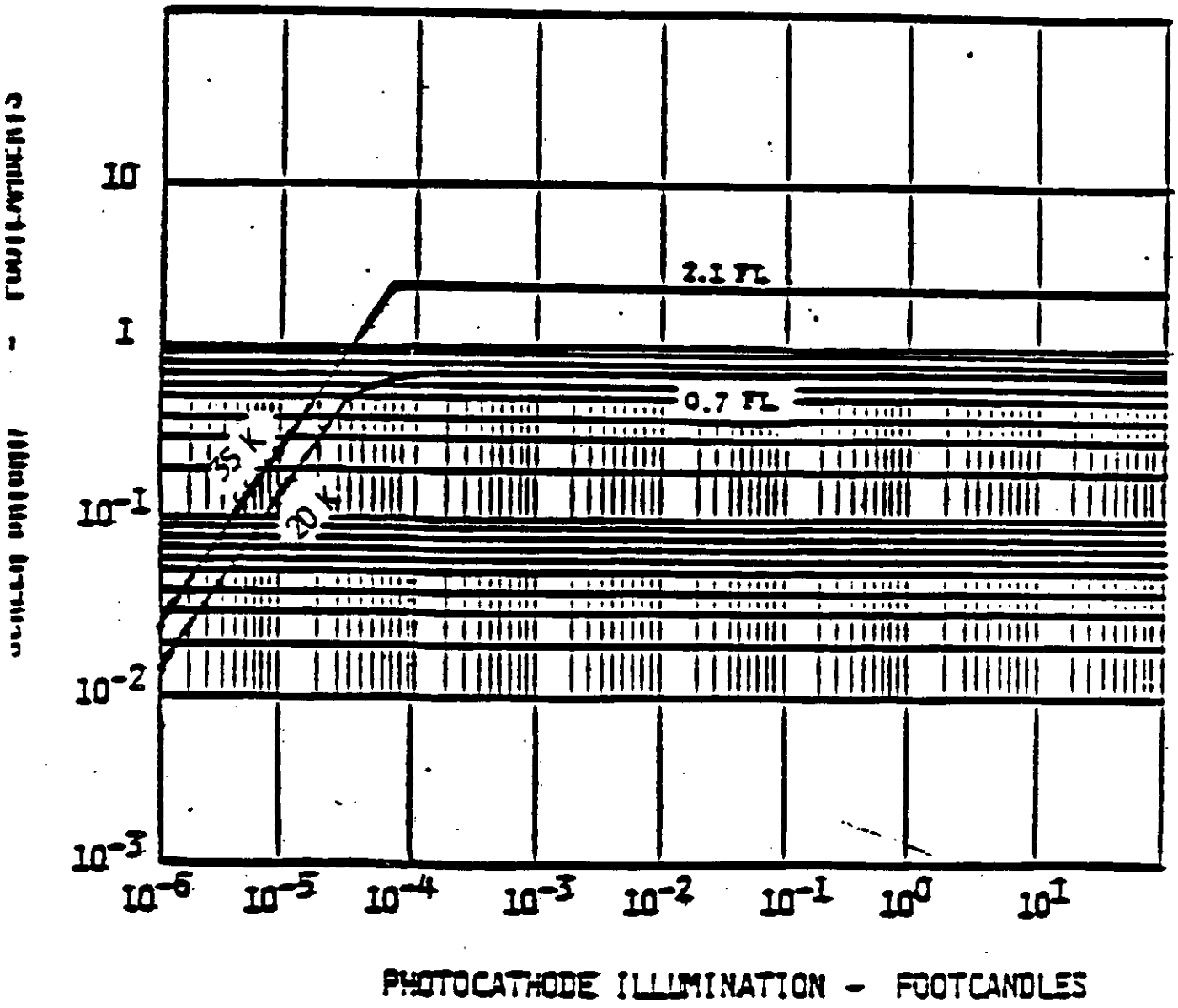


FIG. 3 GAIN/SATURATION REQUIREMENT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-49453	Image Intensifier Assy, 18 MM Microchannel Wafer MX-10130( )/UV

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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